

Hasnat Rasool

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TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML/CSS, Solidity

Frameworks: n8n, Flask, React.js, Angular, Node.js, .NET, Streamlit, Electron.js, LangChain, Web3.js,

Databases: MongoDB, MS SQL Server, PostgreSQL, VectorDB

Tools & AI/ML: Git/GitHub, AWS, Docker, REST APIs, RAG Systems, Ollama, Evolution API, Workflow Automation

EDUCATION

Bachelor of Science in Computer Science

National University of Computer and Emerging Sciences (FAST)

2020 – 2024

Islamabad, Pakistan

EXPERIENCE

Full-Stack Software Developer

Angularspring, Powersoft19

Jan 2025 – Present

Lahore, Pakistan

- Architected a highly concurrent Flask microservice processing 4M+ products, featuring a dual-pipeline LLM extraction engine (Qwen2.5) with strict Pydantic schema enforcement, aggressive regex noise cancellation, and algorithmic anti-hallucination gates.
- Engineered a high-speed classification waterfall utilizing C++ RapidFuzz for MS SQL matching, ensuring WSGI multi-worker thread safety via OS-level semaphores and sub-millisecond latency via SHA-256 semantic MongoDB caching.
- Engineered RAG-based chatbots using LangChain and Ollama (Llama 3.2) with ChromaDB vector databases for context-aware technical assistance and conversation memory management.
- Architected a high-concurrency cross-platform synchronization engine using n8n and Evolution API, featuring a "Transparent Proxy" relay and custom "Loop Killer" logic to handle complex binary media pipelines with sub-100ms response latency.
- Developed an end-to-end bidirectional task management ecosystem integrating WhatsApp and Smartsheet, utilizing custom JavaScript parsing to synchronize real-time employee performance metrics into executive dashboards.
- Engineered ETL pipelines for automated data extraction (web scraping), transformation (Deterministic LLM extraction engine), and loading into SQL Server, with in-memory caching reducing API latency from 2s to 200ms.
- Developed an automated web scraping system using Selenium and BeautifulSoup for extracting ingredient data with pagination handling, retry logic, and structured CSV export for database import.
- Collaborated with a 3-person development team using Git workflows, conducting code reviews, and participating in daily standups for feature development.
- Designed Angular applications for time-series visualization using ECharts WebGL, handling 1M+ data points with real-time interaction, custom timestamp parsing, and CSV export functionality.
- Built desktop applications using Electron.js and .NET (C# WinForms) for internal data management, image processing tools, and cross-platform utilities with secure IPC communication.

Web Development Intern

CyberSoft Vantage

Jul 2023 – Oct 2024

Islamabad, Pakistan

- Developed production MERN application features, including implementing React components with hooks and state management and collaborating on Express.js REST APIs with MongoDB integration.
- Participated in Git-based workflows, including code reviews, pull requests, and CI/CD pipelines, ensuring code quality and deployment standards.

Teaching Assistant

National University of Computer and Emerging Sciences

Spring 2023 – Fall 2023

Islamabad, Pakistan

- Conducted lab sessions for **CS 118**, **CS 1004**, and **CS 2001**, mentoring 100+ students in C/C++ programming, data structures, and debugging techniques.

Event Manager

TEDxFASTIslamabad

Sept 2022 – Dec 2022

Islamabad, Pakistan

- Managed operations and logistics for FAST's first official TEDx event, coordinating speakers, venue setup, and audience engagement initiatives.

PROJECTS

FYP - TrustIsMust | MERN, Python, Machine Learning, AWS

- Developed a full-stack system using MERN with ML integration to detect fake reviews, implementing Naive Bayes classification and PyABSA for aspect-based sentiment analysis using NLP techniques.

- Built an automated review scraping pipeline with multiple ML classifiers generating real-time trust scores, deployed on AWS (EC2, S3, RDS) with MongoDB for high-volume processing.

EngineerBot - AI Technical Assistant | *Python, LangChain, Ollama, RAG, ChromaDB*

- Developed an RAG chatbot using LangChain and Ollama (Llama 3.2) with ChromaDB vector database and mxbai-embed-large embeddings, achieving 95%+ relevance in retrieving engineering design notes from 1,000+ documents using top-k similarity search.
- Built a Streamlit interface with conversation memory, structured data extraction, and an analytics dashboard tracking user interactions and query patterns.

CSV Time Series Visualization Tool | *Angular, TypeScript, ECharts WebGL*

- Built a high-performance Angular app visualizing 1M+ data points with ECharts WebGL, implementing overlay/stacked modes, dynamic channel selection, and custom timestamp parsing (DD-MM-YYYY HH:MM:SS.ms) with microsecond precision.
- Developed a robust CSV parser using PapaParse with row-by-row streaming and progress tracking, processing 500MB+ datasets without browser crashes while maintaining a responsive UI.

CERTIFICATIONS

- AWS Academy Graduate: AWS Academy Microservices and CI/CD Pipeline Builder
- freeCodeCamp: JavaScript Algorithms and Data Structures
- W3 School: HTML and CSS
- Datacamp: Introduction to Python